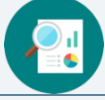
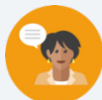


8.6 Solving Real-World Problems With Quadrilaterals

Lesson Introduction

 Benchmarks	MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms. MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.			 Learning Targets	- I can use properties of parallelograms to solve real-world problems. - I can use properties of trapezoids to solve real-world problems.
 Benchmark Coherence	Building On N/A			Working Toward N/A	
 Essential Question	- How can the properties of parallelograms be used to solve real-world problems? - How can the properties of trapezoids be used to solve real-world problems?				
 Lesson Narrative	This lesson brings together the properties and relationships learned in Lessons 1-5 about parallelograms and trapezoids to solve real-world problems.				
 Component Summary	8.6.1 Warm-Up (~5 min.)	Explain errors found in student samples. (SE p. 39)	8.6.3 Your Turn! (5-10 min.)	Apply knowledge of quadrilaterals. (SE p. 41)	
	8.6.2 Activity (35-40 min.)	Solve real-world problems using parallelograms. (SE p. 40)	8.6.4 Wrap-Up (~5 min.)	Reflect on understanding of the properties covered thus far covered in from the unit. (SE p. 42)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Glossary:</u> - N/A		(Optional) Chart paper (Optional) Markers		- This lesson does include a Gallery Walk that needs to be prepared. Consider making additional sets of the problems for large classes. - If using task cards, consider laminating for use in multiple classes. - Consider strategic grouping for 8.6.2 Activity, prior to the start of the lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may set the wrong expressions equal when creating equations to solve.	- For 8.6.2 Activity, consider displaying anchor charts that show the classifications of quadrilaterals, or that show the properties covered thus far in the unit. If students have made a foldable, allow them to refer to it during the activity. - If time is an issue, then consider assigning 8.6.3 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns 8.6.2 Activity is a collaborative Gallery Walk. Students should take turns discussing their work to be able to come to a consensus on each answer. Consider having students alternate who is writing the final answer, but each student should take turns explaining their thinking to allow for both students to have a voice, prior to finalizing their collaborative answer.	MA.K12.MTR.7.1: Real-World Contexts 8.6.2 Activity students are expected to apply the properties learned from the unit to real-world contexts. While students may not initially think about quadrilaterals in the real-world when solving problems, this activity will heighten the awareness of how properties and theorems can be used to solve problems for real-world contexts.
 Differentiate Instruction	Use observational data from prior lessons to determine if there needs to be a teacher-facilitated group for the 8.6.2 Activity. If there is a teacher-facilitated group, keep it small and ensure gradual release occurs during the activity to position students to be more confident in the content by the end of the activity.	
Lesson Notes:		

8.6.1 Warm-Up: Error Analysis

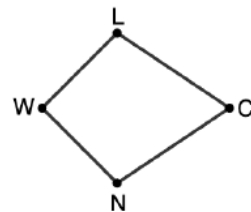
Component Overview: Students review the work of two students that have both made mistakes, and write a letter to each student explaining the mistakes made.



8.6.1 Warm-Up: Error Analysis

Kite $WLCN$ is shown, with $m\angle W = 90^\circ$, $m\angle C = 68^\circ$, and $m\angle N = (6x + 5)^\circ$.

Marcia and Eugene both worked to determine the value of x . Their work is shown.



Marcia	Eugene
$90 + 90 + 68 + 6x + 5 = 360$ $6x + 5 + 248 = 360$ $6x + 253 = 360$ $6x = 107$ $x = \frac{107}{6}$	$360 - (90 + 68) = 202$ $6x + 5 = 202$ $6x = 197$ $x = \frac{197}{6}$



This is an opportunity for students to practice coaching peers, and not just give the answer to students who have made mistakes. Consider allowing students to share their letters and revise based on their partner's feedback.

- Evaluate each student's work, then write a note to each student explaining the error they made.

Marcia	Eugene
<i>Marcia,</i> It looks like you assumed the $m\angle L = 90^\circ$. The measure of $\angle L$ will be equal to $m\angle N$ so you could double that and try to solve with your work $= 360^\circ$. Try it and let me know what you get!	<i>Eugene,</i> It looks like you forgot the measure of angle $\angle L$ when setting up your equation on the second line. The measure of $\angle L$ will be equal to $m\angle N$ so you could double that and try to solve with your work $= 360^\circ$. Try it and let me know what you get!

8.6.2 Activity: World Problem Gallery Walk

Component Overview: Students will engage with a series of real-world problems, and decide what properties of parallelograms and trapezoids are needed to solve each. Students will collaborate on their work to come to a consensus on their answers. Circulate among the groups to ensure that one student is not doing all of the work.



8.6.2 Activity: Word Problem Gallery Walk

1. Work through the word problems with your partner. Show your work and write your answers in the appropriate boxes below.

Problem Number	Work	Answer
1	$KR = DH$ $\frac{1}{2}x + 5 = 2x - 4$ $6 = x$ $KR = \frac{1}{2}(6) + 5 = 8 \text{ in.}$	$KR = 8 \text{ in.}$
2	$DW = NF$ $5y - 5 = 2y + 4$ $y = 3$ $DN = WF$ $12x + 4 = 20x - 8$ $x = 1.5$ $\text{Area} = (5(3) - 5)(12(1.5) + 4)$ $\text{Area} = 220 \text{ ft}^2$	$\text{Area} = 220 \text{ ft}^2$
3	$m\angle MGS = m\angle PSG$ $(8x - 10)^\circ = (6x + 12)^\circ$ $x = 11$ $m\angle MGS = (8(11) - 10)^\circ = 78^\circ$ $m\angle PSG = (6(11) + 12)^\circ = 78^\circ$ $m\angle GMP = m\angle MPS$ $(22y - 19)^\circ = (16y + 14)^\circ$ $y = 5.5$ $m\angle GMP = (22(5.5) - 19)^\circ = 102^\circ$ $m\angle MPS = (16(5.5) + 14)^\circ = 102^\circ$	$m\angle MGS = 78^\circ$ $m\angle PSG = 78^\circ$ $m\angle GMP = 102^\circ$ $m\angle MPS = 102^\circ$



Students should take turns explaining how to approach the problem and/or take turns in sharing their work. Then, students should come to a consensus on the correct answer(s).

8.6.2 Activity: World Problem Gallery Walk (continued)

4	$m\angle PCH = m\angle PTH$ $(6x - 23)^\circ = (4x + 19)^\circ$ $x = 21$ $m\angle PCH = (6(21) - 23)^\circ = 103^\circ$ $2(m\angle PCH)^\circ + 2(m\angle CPT)^\circ = 360^\circ$ $2(103)^\circ + 2(8y - 23)^\circ = 360^\circ$ $y = 12.5$	$y = 12.5$
5	$FM = MA$ $3x + 20 = 5x - 3$ $x = 11.5$ $FM = 3(11.5) + 20 = 54.5$ $RM = \frac{1}{2}(FL)$ $RM = \frac{1}{2}(8x - 1)$ $RM = \frac{1}{2}(8(11.5) - 1) = 45.5$ $TM = \frac{1}{2}(FG)$ $TM = \frac{1}{2}(5x + 2.5)$ $TM = \frac{1}{2}(5(11.5) + 2.5) = 30$	$FM = 54.5 \text{ in.}$ $RM = 45.5 \text{ in.}$ $TM = 30 \text{ in.}$
6	$m\angle DMF + m\angle MDW + m\angle DWF$ $+ m\angle WFM = 360^\circ$ $(13x + 4.4)^\circ + (10x + 13)^\circ + (15x + 4)^\circ + (10x - 7)^\circ = 360^\circ$ $x = 7.2$ $m\angle DMF = (13(7.2) + 4.4)^\circ = 98^\circ$ $m\angle DWF = (15(7.2) + 4)^\circ = 112^\circ$ $m\angle MDW = (10(7.2) + 13)^\circ = 85^\circ$ $m\angle WFM = (10(7.2) - 7)^\circ = 65^\circ$ <i>Because opposite angles are not congruent, this is not a parallelogram.</i>	<i>DWFM is not a parallelogram.</i>



Consider one of the following options for this activity.

- **Gallery Walk** – hang problems around the room. Groups are assigned to work one problem on chart paper and post. When all groups have finished, the groups rotate from problem to problem, verifying and editing any mistakes.
- **Task Cards** – print the problems on separate pieces of paper. Give each group one card to complete. When finished, pass the card to the next group. Continue until all cards have been completed.

For large groups, consider making two sets of problems.

8.6.2 Activity: World Problem Gallery Walk (continued)

7	$HX = 5x - 2$ $WR = 2 \cdot HX$ and $WR = 9x + 3$ $9x + 3 = 2(5x - 2)$ $7 = x$ $KR = 11x + 19$ $KR = 11(7) + 19 = 96 \text{ cm}$ $WR = 9(7) + 3 = 66 \text{ cm}$	The dimensions of $KRWF$ are $96 \text{ cm} \times 66 \text{ cm}$
8	$XW = DN$ $12x - 15 = 9x + 27$ $x = 14$ $DW = XN$ $DW = (8x + 40.5) \text{ m}$ $DW = (8(14) + 40.5) \text{ m}$ $DW = 152.5 \text{ m}$	$DW = 152.5 \text{ m}$



When all pairs are finished, the suggested activity is to put pairs in larger groups to compare answers, discuss any differences, and make corrections. Then, larger groups write final answers on chart paper designated for each problem. At that point, discuss any differences as a class, and work together to correct mistakes.

8.6.3 Your Turn!

Component Overview: Students respond to prompts about quadrilaterals that allow them to analyze each type of quadrilateral they have learned about in prior lessons. Students will have to select quadrilaterals that fulfill a condition, then offer a description of a given quadrilateral.



8.6.3 Your Turn!

- I am a quadrilateral with two pairs of opposite parallel sides. At least one of my angles has a measure of 90° and both of my diagonals measure 12 inches. Select all shapes that match this description.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite
- In your own words, write a description of a parallelogram that could **only** match a square.
A square is a quadrilateral with four equal sides, four right angles, and diagonals that are congruent.
- If a quadrilateral is a parallelogram, what additional information would be needed to show that the shape is a rectangle?
To show a parallelogram is a rectangle, it would need to have four right angles and diagonals that are congruent.
- If a quadrilateral is a parallelogram, what additional information would be needed to show that the shape is a rhombus?
To show a parallelogram is a rhombus, it would need to have four congruent sides.



If time is an issue, then consider assigning 8.6.3 Your Turn! for homework.



Consider allowing students options for responding to questions 1-3. For example, instead of students writing their answers, allow them to choose to explain their thinking verbally to a partner who they haven't worked with so far in the lesson.

8.6.4 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of properties and completing proofs of parallelograms, rectangles, squares, rhombi, kites, and trapezoids.



8.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.



I need help.



I got it.



I could help someone.

Answers will vary

Properties of a parallelogram			
Properties of a rectangle			
Properties of a kite			
Completing proofs using properties and theorems about parallelograms			
Using the properties of a parallelogram to solve problems			
Properties of a trapezoid			
Using the properties of a trapezoid to solve problems			



Make note of students who mark any statements as “I need help.” After the lesson, determine a plan for how to provide support to those students (i.e., strategic grouping during the next lesson or during the end of unit review, pull small groups, etc.).